

**JOMO KENYATTA UNIVERSITY OF
AGRICULTURE AND TECHNOLOGY**

**DEPARTMENT OF PURE AND APPLIED
MATHEMATICS**

**SMA 2304: ORDINARY DIFFERENTIAL
EQUATIONS 1**

CLASS NOTES

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SMA 2304

Ordinary Differential Equations 1
SMA 2304

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CHAPTER 1

INTRODUCTION

1.1 Course Outline

Evaluation

* Course work:
30%

- (a) Two constants
- (b) Assignment.

Final Examination: 70%

- + *Definitions: differential equations, order and degree, linearity origin of differential equation and solutions to differential equations.*
- + *First order first degree equations, definition, exact, variable separable, homogeneous, linear equation of first order (Bernoulli) and Bernoulli equation.*
- + *Cat 1*
- + *Higher order differential equations: Homogeneous equations with constant coefficients. Non-homogeneous equation with constant coefficients. The method of variation of parameters, the method of undetermined coefficients*
- + *Assignment*
- + *Systems of ordinary differential equations.*
- + *Equations with variable coefficients. Power series, The Cauchy-Euler equations, Solutions to ordinary differential Equations, Taylor series solutions and general power series about an ordinary point and a regular singular point.*
- + *Cat 2*

References

1. *Differential equations by Frank Ayrox*
2. *Engineering mathematics by Edwin Kreyezig*

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CHAPTER 2

DEFINITIONS

A differential equation is an equation involving derivatives of a function and / or the function itself:

e.g.,

$$\frac{dy}{dx} + 4x = 2 \quad (2.1)$$

$$\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 35y = 0 \quad (2.2)$$

$$\frac{\partial y}{\partial t} = 2\frac{d^2y}{dx^2} \quad (2.3)$$

Differential equations can be classified into two, namely:

Definition 1. Ordinary Differential equations: This is an equation involving derivatives of a function with respect to only one independent variable thus the derivatives involved are complete derivatives. $\left(\frac{dy}{dx}\right)$.

Ex equation 2.1 and 2.2 are examples of ordinary differential equation.

Definition 2. Partial Differential Equation. This is an equation involving derivatives of a function with respect to more than one independent variables thus the derivatives involved are partial. $\left(\frac{\partial}{\partial x}\right)$

Definition 3. The order of a differential equation

This is the order of the highest order derivative appearing in a differential equation.

E.g, Equation 2.1 is order 1 while equation 2.2 and 2.3 are of order 2.

Definition 4. Degree of a differential Equation

This is the degree of the highest order derivative appearing in a differential equation.

Eg, equation 2.1 to 2.3 are of degree 1.

Quiz 1

Fill in the following table.

Equation	Order	Degree
$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 = 4x$	2	3
$\frac{dy}{dx} = 2e^x$	1	1
$\frac{d^4y}{dx^4} + 4\left(\frac{d^3y}{dx^3}\right)^5 + 4y^6 = 0$	4	1

2.1 Linear and non linear differential equations

1

Consider the following:

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + a_2(x)\frac{d^{n-2}y}{dx^{n-2}} + \dots + a_{n-2}(x)\frac{d^2y}{dx^2} + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = F(x) \quad (2.4)$$

NB: Equation 2.4 is of order n degree 1. We also note that the function y and all its derivatives are of first degree and that here are no products of y with any of its derivatives. Such an equation is said to be linear in y . Otherwise the equation is said to be non linear.

Example: The following equations are linear in y

a.

$$\frac{dy}{dx} + 4y = 2x$$

b.

$$\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 35y = 4 \ln x$$

¹**Difference between Linear and Nonlinear Differential Equation** A differential equation, which has only the linear terms of the unknown or dependent variable and its derivatives, is known as a linear differential equation. It has no term with the dependent variable of index higher than 1 and do not contain any multiple of its derivatives. It cannot have nonlinear functions such as trigonometric functions, exponential function, and logarithmic functions with respect to the dependent variable. Any differential equation that contains above mentioned terms is a nonlinear differential equation.

And the following equations are non linear in y

a.

$$\frac{dy}{dx} + y^2 = 2x$$

b.

$$\left(\frac{d^2y}{dx^2}\right)^3 + 4\frac{dy}{dx} + 2y = 0$$

c.

$$\frac{d^2y}{dx^2} + 4y\frac{dy}{dx} + 2x = e^x$$

CHAPTER 3

FORMATION OF ORDINARY DIFFERENTIAL EQUATIONS

Differential equations may arise from either :

- i. A physical problem
- ii. A geometric problem *or*
- iii. A primitive problem.

A primitive is a relation between the dependent variable (y) and the independent variable (x) involving n -essential arbitrary constants.

Arbitrary constants are said to be essential if they cannot be replaced by a smaller number of constants. Therefore, a primitive with one constant, will result to a differential equation of order 1 while a primitive with 2 constants will result to a differential equation of order 2 *e.t.c*

Example

Determine differential equation arising from

$$y = c_1 e^{5x} + c_2 e^{7x} \quad (3.1)$$

Task: Differentiate twice and eliminate the constants c_1 and c_2 to form the ODE. Therefore, differentiating 3.1 with respect to x

$$\frac{dy}{dx} = 5c_1 e^{5x} + 7c_2 e^{7x} \quad (3.2)$$

$$\frac{d^2y}{dx^2} = 25c_1 e^{5x} + 49c_2 e^{7x} \quad (3.3)$$

Multiplying equation 3.1 by 5 gives

$$5y = 5c_1e^{5x} + 5c_2e^{7x} \quad (3.4)$$

Then subtract 3.4 from 3.2

$$\frac{dy}{dx} - 5y = 2c_2e^{7x} \quad (3.5)$$

Also multiplying equation 3.1 by 25 gives

$$25y = 25c_1e^{5x} + 25c_2e^{7x} \quad (3.6)$$

Then subtract 3.6 from 3.3

Use 3.5 in 3.6 to have

$$\frac{d^2y}{dx^2} - 25y = 12 \left(\frac{dy}{dx} - 5y \right)$$

or

$$\frac{d^2y}{dx^2} - 25y = 12 \frac{dy}{dx} - 60y$$

This gives

$$\frac{d^2y}{dx^2} - 12 \frac{dy}{dx} + 35 = 0 \quad (3.7)$$

CHAPTER 4

SOLUTION TO A DIFFERENTIAL EQUATION

This is a function which together with its derivatives satisfy a given differential equation.

Example

Show that

$$y = c_1 e^{5x} + c_2 e^{7x} \quad (4.1)$$

is a solution to

$$\frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 35y = 0 \quad (4.2)$$

Solution:

Task: Differentiate the primitive twice and substitute into the given differential equation

$$\frac{dy}{dx} = 5c_1 e^{5x} + 7c_2 e^{7x} \quad (4.3)$$

$$\frac{d^2 y}{dx^2} = 25c_1 e^{5x} + 49c_2 e^{7x} \quad (4.4)$$

Substituting 4.1 , 4.3 and 4.4 in 4.2 gives

$$25c_1 e^{5x} + 49c_2 e^{7x} - 12(5c_1 e^{5x} + 7c_2 e^{7x}) + 35(c_1 e^{5x} + c_2 e^{7x}) = \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 35y$$

$$25c_1 e^{5x} + 49c_2 e^{7x} - 60c_1 e^{5x} - 84c_2 e^{7x} + 35c_1 e^{5x} + 35c_2 e^{7x} = \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 35y$$

Hence

$$\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 35y = 0$$

, thus a solution.

Also show that $y_1 = e^{5x}$ and $y_2 = e^{7x}$ are solutions

$$\begin{aligned} y_1' &= 5e^{5x} \\ y_1'' &= 25e^{5x} \\ y_1'' - 12y_1' + 35y_1 &= 25e^{5x} - 12(5e^{5x}) + 35e^{5x} \\ &= 25e^{5x} - 60e^{5x} + 35e^{5x} \\ &= 0 \end{aligned}$$

Hence a solution

$$\begin{aligned} y_2' &= 7e^{7x} \\ y_2'' &= 49e^{7x} \\ y_2'' - 12y_2' + 35y_2 &= 49e^{7x} - 12(7e^{7x}) + 35e^{7x} \\ &= 49e^{7x} - 84e^{7x} + 35e^{7x} \\ &= 0 \end{aligned}$$

Hence a solution

CHAPTER 5

FIRST ORDER FIRST DEGREE EQUATIONS

A first order first degree equation, is an equation of the form

$$\frac{dy}{dx} = f(x, y) \quad (5.1)$$

E.g.,

$$\frac{dy}{dx} = -\frac{(x^2 + y^2)}{2xy}$$

Which can also be written as

$$(x^2 + y^2)dx + 2xydy = 0$$

Which is of the form

$$M(x, y)dx + N(x, y)dy = 0$$

Or simply

$$Mdx + Ndy = 0 \quad (5.2)$$

5.1 Exact Equations

The differential equations $M(x, y)dx + N(x, y)dy = 0$ is said to be exact if there exists a two valued function $F(x, y)$ such that the total differential of F is equal to this equation. That is, $\frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = Mdx + Ndy$ ¹ Equating coefficients of dx and dy gives

$$\frac{\partial F}{\partial x} = M \quad (5.3)$$

¹Total Differential $F(x, y)$

$$dF = \frac{\partial F}{\partial x}dx$$

$$\frac{\partial F}{\partial y} = N \quad (5.4)$$

We now differentiate equation 5.3 w.r.t y and equation 5.4 partially w.r.t x to have

$$\frac{\partial^2 F}{\partial y \partial x} = \frac{\partial M}{\partial y} \quad (5.5)$$

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial N}{\partial x} \quad (5.6)$$

By the continuity of second order partial derivatives, we know that

$$\frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y}$$

Then it follows that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (5.7)$$

This is test for exactness.

By definition of an exact equation, we have $dF = 0$ ²

Integrating this, gives $\int dF = \int 0$ or $F = c$

Example 1

Determine whether the equations

$$(x^2 + y^2)dx + 2xydy = 0$$

is exact, hence find its solution.

Solution

By definition

$$M = x^2 + y^2 \implies \frac{\partial M}{\partial y} = 2y$$

and

$$N = 2xy \implies \frac{\partial N}{\partial x} = 2y$$

Hence the equation is exact.

²

$$Mdx + Ndy = 0$$

$$\frac{\partial F}{\partial x} dy \dots$$

From equations 5.1 and 5.2 we have

$$\frac{\partial F}{\partial x} = x^2 + y^2 \dots (A) \quad (5.8)$$

$$\frac{\partial F}{\partial y} = 2xy \dots (B) \quad (5.9)$$

Integrating equation 5.8 partially, gives , w.r.t x

$$F = \frac{x^3}{3} + xy^2 + \Phi(y) \dots (C) \quad (5.10)$$

Differentiating equation 5.10 partially, w.r.t y,gives,

$$\frac{\partial F}{\partial y} = 2xy + \Phi'(y) \dots (D) \quad (5.11)$$

Where

$$\Phi'(y) = \frac{d\Phi(y)}{dy}$$

Equating D and B we have

$$2xy + \Phi'(y) = 2xy$$

$$\Phi'(y) = 0$$

Which on integration gievs

$$\Phi(y) = k, \text{ a constant}$$

$\therefore$

$$F = \frac{x^3}{3} + xy^2 + k$$

and the required solution is

$$F = C$$

$$\frac{x^3}{3} + xy^2 + k = c$$

Or

$$\frac{x^3}{3} + xy^2 = c_1$$

where $c_1 = c - k$

Check

$$\frac{x^3}{3} + xy^2 = c_1$$

Differentiate

$$3\frac{x^2}{3}dx + y^2dx + 2xydy = 0$$

$$(x^2 + y^2)dx + 2xydy = 0$$

Alternative 1

We have A and B

Intergrate B with respect to y

$$F = \frac{2xy^2}{2} + \Phi(x)$$

$$F = xy^2 + \Phi(x) \quad (5.12)$$

$$\frac{\partial F}{\partial x} = y^2 + \Phi'(x) \quad (5.13)$$

Equate D and A

$$\begin{aligned} y^2 + \Phi'(x) &= x^2 + y^2 \\ \Phi'(x) &= x^2 \end{aligned}$$

Which on integration gives

$$\Phi(x) = \frac{x^3}{3} = k \text{ a constant}$$

Substitute $\Phi(x)$ in C to have

$$F = xy^2 + \frac{x^3}{3} + k$$

Alternative 2

Grouping method

In this method, we try to group some terms into total differentials of either products or quotients by inspection then integrate to obtain the required solution.

E.g., given

$$(x^2 + y^2)dx + 2xydy = 0$$

Opening the bracket gives,

$$x^2dx + y^2dx + 2xydy = 0$$

By inspection, the second and third terms can be combined to $d(xy^2)$. Therefore we have ³

$$d\left(\frac{x^3}{3} + d(xy^2)\right) = d(c)$$

which on integration gives

$$\frac{x^3}{3} + xy^2 = c$$

³ $d(c)$ Derivative of a constant is equal to zero.

5.2 Variable Separable Equations

A first order first degree equation is said to be variable separable if it can be written in the form

$$g_1(x)g_2(y)dx + g_3(x)g_4(y)dy = 0 \quad (5.14)$$

Dividing equation 5.14 by $g_2(y)g_3(x)$ gives

$$\frac{g_1(x)g_2(y)dx}{g_2(y)g_3(x)} + \frac{g_3(x)g_4(y)dy}{g_2(y)g_3(x)} = 0$$

or

$$\frac{g_1(x)}{g_3(x)}dx + \frac{g_4(y)}{g_2(y)}dy = 0$$

This is of the form

$$M(x)dx + N(y)dy = 0 \quad (5.15)$$

Equation 5.15 is exact since

$$\frac{\partial M(x)}{\partial y} = \frac{\partial N(y)}{\partial x} = 0$$

And the solution to equation 5.15 is

$$\int M(x)dx + \int N(y)dy = C$$

Example

Determine the solution to the equation

$$x \sin y dx + (1 + x^2) \cos y dy = 0$$

Solution:

Divide through by $(1 + x^2) \sin y$

$$\frac{x}{1 + x^2}dx + \frac{\cos y}{\sin y}dy = 0$$

Then integrate

$$\int \frac{x}{1 + x^2}dx + \int \frac{\cos y}{\sin y}dy = c$$

or

$$\frac{1}{2} \ln(1 + x^2) + \ln(\sin y) = c$$

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⁴ $\int \frac{x}{1+x^2} dx, u = 1 + x^2, \frac{du}{dx} = 2x$

Example

The number of words N per minute that a first year student can type in a computer increases with practice. The rate of change of N with time t (days of practice) is proportional to the difference between N and an upper limit of 140 words per minute. At the beginning of the year, the student could not type at all, and after 10 days of practice, the student could type 35 words per minute.

- i) Form a differential equation relating N and t and give a condition that N must satisfy.
- ii) How many words per minute will the student be able to type after 20 days of practice?
- iii) How long must a student practice to be able to type 105 words per minute?

Solution

i)

$$\begin{aligned}\frac{dN}{dt} &\propto (140 - N) \\ \frac{dN}{dt} &= k(140 - N)\end{aligned}$$

Where k is a constant of proportionality. Condition $0 \leq N \leq 140$

ii)

$$\begin{aligned}\frac{dN}{dt} &= k(140 - N) \\ \frac{dN}{140 - N} &= k dt \text{ separate variable} \\ \int \frac{dN}{140 - N} &= k \int dt\end{aligned}$$

Or

$$-\ln(140 - N) = kt + C \text{ where } C \text{ is a constant of integration}$$

$$\ln(140 - N) = -kt + B, B = -C$$

$$e^{\ln(140 - N)} = e^{-kt + B}$$

$$140 - N = e^{-kt} \cdot e^B$$

$$140 - N = Ae^{-kt}, A = e^B$$

$$\implies N = 140 - Ae^{-kt}$$

at $t = 0, N = 0$, i.e., at the beginning, the student could type 0 words per minute.

$$0 = 140 - A$$

$$A = 140$$

$$\implies N = 140 - 140e^{-kt}$$

$$N = 140(1 - e^{-kt})$$

$$t = 10, N = 35$$

$$35 = 140(1 - e^{-10k})$$

$$\frac{35}{140} = 1 - e^{-10k}$$

$$0.25 = 1 - e^{-10k}$$

$$-0.75 = e^{-10k}$$

$$e^{-10k} = 0.75$$

$$-10k = \ln(0.75)$$

$$k = \frac{\ln(0.75)}{-10}$$

$$k = 0.0287682$$

$$\therefore N = 140(1 - e^{-0.0287682t})$$

Hence

$$N(20) = 140 [1 - e^{-0.0287682(20)}]$$

$$= 61.25$$

$$\approx 61 \text{ words}$$

$$\text{iii) } 105 = 140(1 - e^{-0.0287682t})$$

$$\frac{105}{140} = 1 - e^{-0.0287682t}$$

$$0.75 = 1 - e^{-0.0287682t}$$

$$0.25 = -e^{-0.0287682t}$$

$$e^{-0.0287682t} = 0.25$$

$$-0.0287682t = \ln(0.25)$$

$$t = \frac{\ln(0.25)}{-0.0287682} = 48.18$$

$$\approx 49 \text{ days}$$

5.2.1 Integrating Factors

Consider the equation $M(x, y)dx + N(x, y)dy = 0$

If $\frac{\partial M(xy)}{\partial y} = \frac{\partial N(xy)}{\partial x}$ then the equation is exact, the solution of which is one parameter family of solutions. If $\frac{\partial M(xy)}{\partial y} \neq \frac{\partial N(xy)}{\partial x}$ the equation is not exact thus the above method cannot be applied to solve the equation.

Perhaps this equation can be multiplied by an expression which will transform it to an exact one. e.g., Consider the equation.

$$ydx + 2xdy = 0$$

, this equation is not exact but if multiplied by y it is transformed to an exact one

$$y^2dx + 2xydy = 0$$

Since this equation is integrable y is called the integrating factor.

Definition 5. If the equation

$$M(x, y)dx + N(x, y)dy = 0$$

is not exact in domain D but the differential equation

$$\mu(xy)M(xy)dx + \mu(xy)N(xy)dy = 0$$

is exact in D then $\mu(xy)$ is called an integrating factor of the given equation.

Example:

Consider the differential equation $(3y + 4xy^2)dx + (2x + 3x^2y)dy = 0$

Now $\frac{\partial M}{\partial y} = 3 + 8xy$ and $\frac{\partial N}{\partial x} = 2 + 6xy$. This equation is not exact. Let $\mu(xy) = x^2y$ then multiply the given equation by $\mu(xy)$ we have $(3x^2y^2 + 4x^3y^2)dx + (2x^2y + 3x^4y^2)dy$ which is exact since

$$\frac{\partial M}{\partial y}(xy)\mu = 6x^2y + 12x^3y^2$$

$$\frac{\partial N}{\partial x}\mu(xy) = 6x^2y + 12x^3y^2$$

hence $\mu(xy)$ is the integrating factor of this equation.

5.3 Homogeneous Equations

The equation

$$M(x, y)dx + N(x, y)dy = 0 \tag{5.16}$$

is said to be homogeneous if it can be written in the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right) \quad (5.17)$$

Example:

$$(x^2 + y^2)dx + 2xydy = 0$$

can be written as,

$$2xydy = -(x^2 + y^2)dx$$

or

$$\frac{dy}{dx} = \frac{-(x^2 + y^2)}{2xy}$$

or

$$\frac{dy}{dx} = \frac{-x^2}{2xy} - \frac{y^2}{2xy}$$

or

$$\frac{dy}{dx} = -\frac{1}{2} \frac{x}{y} - \frac{1}{2} \frac{y}{x}$$

or

$$\frac{dy}{dx} = -\frac{1}{2} \frac{1}{\frac{y}{x}} - \frac{1}{2} \frac{y}{x}$$

Which is of the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

5.3.1 Test for Homogeneity

Definition 6. A function $F(x, y)$ is said to be homogeneous degree n if on replacing x with tx and y with ty we have $F(tx, ty) = t^n F(x, y)$

Eg.

$$M(x, y) = x^2 + y^2$$

$$\begin{aligned} M(tx, ty) &= (tx)^2 + (ty)^2 \\ &= t^2x^2 + t^2y^2 \\ &= t^2(x^2 + y^2) \\ &= t^2M(x, y) \end{aligned}$$

Hence homogeneous degree 2

Also $N(x, y) = 2(xy)$

$$\begin{aligned} N(tx, ty) &= 2(tx)(ty) \\ &= 2t^2xy \\ &= t^2(2xy) \\ &= t^2N(x, y) \end{aligned}$$

Also homogeneous degree 2.

Definition 7. *The differential Equation*

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be homogeneous degree n if both M and N are homogeneous degree n .

Eg.,

$$(x^2 + y^2)dx + 2xydy = 0 \text{ is homogeneous degree 2}$$

Theorem 1: T

The substitution

$$y = vx \tag{5.18}$$

reduced the homogeneous equation

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right) \tag{5.19}$$

into a variable separable equation in v and x

Proof. From 5.18, differentiate with respect to x to have

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \tag{5.20}$$

⁵ Substituting equation 5.18 and 5.20 in 5.19,

$$v + x \frac{dv}{dx} = f(v)$$

or

$$vdx + xdv = f(v)dx$$

or

$$[v - f(v)] dx + xdv = 0$$

Which on separating variables, becomes

$$\frac{dx}{x} + \frac{dv}{v - f(v)} = 0$$

And the required solution is

$$\int \frac{dx}{x} + \int \frac{dv}{v - f(v)} = C$$

⁶

□

⁵Product Rule: $y = uv, y' = u dv + v du$

⁶A constant

Example

Determine the solution to

$$(x^2 + y^2)dx + 2xydy = 0$$

Solution:

This equation is homogeneous degree 2 (Already seen). Then the equation can be written as

$$\frac{dy}{dx} = \frac{-(x^2 + y^2)}{2xy} \dots (A)$$

Let

$$y = vx \longleftrightarrow v = \frac{y}{x} \dots (B)$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \dots (C)$$

Substituting B and C in A gives

$$v + x \frac{dv}{dx} = \frac{-(x^2 + v^2x^2)}{2x(vx)}$$

or

$$v + x \frac{dv}{dx} = \frac{-x^2(1 + v^2)}{x^2(2v)}$$

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or

$$v + x \frac{dv}{dx} = -\frac{1 + v^2}{2v}$$

or

$$vdx + xdv = -\frac{1 + v^2}{2v}dx$$

$$\left[v + \frac{(1 + v^2)}{2v} \right] dx + xdv = 0$$

$$\frac{2v^2 + 1 + v^2}{2v}dx + xdv = 0$$

$$\frac{1 + 3v^2}{2v}dx + xdv = 0$$

$$\frac{dx}{x} + \frac{2v}{1 + 3v^2}dx = 0$$

$$\int \frac{dx}{x} + \int \frac{2v}{1 + 3v^2}dv = \int 0$$

⁷ If it is homogenous degree n then x^n must cancel

or

$$\ln x + \frac{1}{3} \ln(1 + 3x^2) = \ln 3$$

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or

$$\begin{aligned} 3 \ln x + \ln(1 + 3x^2) &= 3 \ln 3 \\ \ln x^3 + \ln(1 + 3x^2) &= \ln B, B = C^3 \\ \ln x^3(1 + 3x^2) &= \ln B \end{aligned}$$

or

$$\begin{aligned} x^3(1 + 3x^2) &= B \\ v &= \frac{y}{x} \\ x^3 \left(1 + \frac{3y^2}{x^2} \right) x^2 &= B \\ x^3 + 3xy^2 &= B \end{aligned}$$

Example

Determine the solution to

$$(y + \sqrt{x^2 - y^2}) dx - x dy = 0$$

Solution:

$$\begin{aligned} M(x, y) &= y + \sqrt{x^2 - y^2} \\ M(tx, ty) &= ty + \sqrt{t^2 x^2 - t^2 y^2} \\ &= ty + \sqrt{t^2 (x^2 - y^2)} \\ &= ty + t\sqrt{x^2 - y^2} \\ &= t \left[y + \sqrt{x^2 - y^2} \right] \end{aligned}$$

Hence, homogeneous degree 1

$$\begin{aligned} N(x, y) &= -x \\ N(tx, ty) &= -tx = t(-x) \end{aligned}$$

$$\begin{aligned} u &= 1 + 3x^2 \\ \frac{du}{dx} &= 6x \\ dx &= \frac{du}{6x} \\ \int \frac{dx}{u} \cdot \frac{du}{6x} &= \frac{1}{3} \int \frac{du}{u} = \frac{1}{3} \ln u \\ &= \frac{1}{3} \ln(1 + 3x^2) \end{aligned}$$

Also homogeneous degree 1.

Let $y = vx \implies v = \frac{y}{x} \dots (A)$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \dots (B)$$

$$\text{From } \frac{dy}{dx} = \frac{y + \sqrt{x^2 - y^2}}{x} \dots (C)$$

Substitue A and B in C .

$$\begin{aligned} v + x \frac{dv}{dx} &= \frac{vx + \sqrt{x^2 - v^2x^2}}{x} \\ v + x \frac{dv}{dx} &= \frac{vx + \sqrt{x^2(a - v^2)}}{x} \\ v + x \frac{dv}{dx} &= \frac{vx + x\sqrt{1 - v^2}}{x} \\ v + x \frac{dv}{dx} &= \frac{x(v + \sqrt{1 - v^2})}{x} \\ v + x \frac{dv}{dx} &= v + \sqrt{1 - v^2} \\ x \frac{dv}{dx} &= \sqrt{1 - v^2} \\ \frac{dv}{1 - v^2} &= \frac{dx}{x} \\ \int \frac{dv}{\sqrt{1 - v^2}} &= \int \frac{dx}{x} \\ \sin^{-1} v &= \ln x = C \end{aligned}$$

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5.4 Lebenitz Equation

This is an equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (5.21)$$

The equation can be written as

$$dy + P(x)ydx = Q(x)dx$$

or

$$^9 \int \frac{dv}{\sqrt{1 - v^2}} = \sin^{-1} v$$

$$[P(x)y - Q(x)] dx + dy = 0 \quad (5.22)$$

For equation 5.22 to be exact, we have

$$\frac{\partial}{\partial y} [P(x)y - Q(x)] = \frac{\partial(1)}{\partial x}$$

or

$$P(x) = 0$$

But we require $P(x) \neq 0$, therefore multiply this equation 5.22 by $\beta(x)$, becomes

$$[\beta(x)P(x)y - \beta(x)Q(x)] dx + \beta(x)dy = 0 \quad (5.23)$$

If 5.23 is exact, we have

$$\frac{\partial}{\partial y} [\beta(x)P(x)y - \beta(x)Q(x)] = \frac{d}{dx} [\beta(x)] \quad (5.24)$$

or

$$\beta(x)P(x) = \frac{d\beta(x)}{dx}$$

or

$$\frac{d\beta(x)}{\beta(x)} = P(x)dx$$

$$\int \frac{d\beta(x)}{\beta(x)} = \int P(x)dx$$

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or

$$\ln \beta(x) = \int P(x)dx$$

Hence

$$\beta(x) = e^{\int P(x)dx} \quad (5.25)$$

Equation 5.25 is known as the intergrating factor¹¹ of equation 5.21.

Example

Determine the solution to the equaiton

$$\frac{dy}{dx} + 2x^3y = 1$$

Solution

¹⁰ $\int \frac{f'(x)}{f(x)} dx = \ln(f(x))$

¹¹ An integrating factor is an a function that makes a non-exact equation to be exact

$$\frac{dy}{dx} + \frac{2x^3y}{x^4} = \frac{1}{x^4}$$

or

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{1}{x^4} \quad (5.26)$$

Integrating factor $P(x) = \frac{2}{x}$

$$\begin{aligned} \beta(x) &= e^{\int \frac{2}{x} dx} = e^{2 \int \frac{dx}{x}} \\ &= e^{2 \ln x} \\ &= e^{\ln x^2} \\ &= x^2 \end{aligned} \quad (5.27)$$

Multiplying equation 5.26 through by the integrating factor, gives

$$x^2 \frac{dy}{dx} + 2xy = \frac{1}{x^2}$$

or

The next step is made up of dependent variable and the independent factor. y is the d.v, x^2 is the I.F

$$\frac{d}{dx} [yx^2] = \frac{1}{x^2}$$

Which on integration gives

$$\begin{aligned} \int \frac{d}{dx} [yx^2] &= \int \frac{1}{x^2} dx \\ yx^2 &= -\frac{1}{x^2} + C \end{aligned}$$

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Example

Determine the solution to

$$(x^2 - 1) \frac{dy}{dx} + 4xy = x$$

Solution:

$$^{12} \int x^{-2} dx = \frac{x^{-2+1}}{-2+1} = -x^{-1} = -\frac{1}{x}$$

The equation can be written as

$$\frac{dy}{dx} + \frac{4x}{x^2 - 1}y = \frac{x}{x^2 - 1} \quad (5.29)$$

$$\begin{aligned} \beta(x) &= e^{\int \frac{4x}{x^2-1} dx} \\ &= e^{2 \ln(x^2-1)} \\ &= e^{\ln(x^2-1)^2} \\ &= (x^2 - 1)^2 \end{aligned} \quad (5.30)$$

Multiplying equation 5.29 through by the intergration factor

$$\begin{aligned} (x^2 - 1)^2 \frac{dy}{dx} + 4x(x^2 - 1)y &= x(x^2 - 1) \\ \frac{d}{dx} [y(x^2 - 1)^2] &= x^3 - x \end{aligned}$$

Proof.

$$\begin{aligned} (x^2 - 1)^2 \frac{dy}{dx} + 2(x^2 - 1)2xy \\ 4xy(x^2 - 1) \end{aligned}$$

□

Then integrate

$$\begin{aligned} y(x^2 - 1)^2 &= \int (x^3 - x) dx \\ y(x^2 - 1)^2 &= \frac{x^4}{4} - \frac{x^2}{2} + C \end{aligned}$$

5.5 The Bernoulli Equation

This is an equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, n \neq 0, 1 \quad (5.31)$$

¹³

Theorem 2: t

The transformation

$$v = y^{1-n} \quad (5.32)$$

reduces the Bernoulli equation 5.32 into a Lebenitz equation in v and x .

¹³ n cannot be 0 or 1. When its 1, it becomes separable, 0, Lebenitz

Proof. differentiate with respect to x to have Form equation 5.32

$$\frac{dv}{dx} = (1 - n)y^{-n} \frac{dy}{dx} \quad (5.33)$$

¹⁴ Or

$$\begin{aligned} \frac{1}{(1 - n)y^{-n}} \frac{dv}{dx} &= \frac{dy}{dx} \\ \frac{y^n}{1 - n} \frac{dv}{dx} &= \frac{dy}{dx} \end{aligned} \quad (5.34)$$

¹⁵ Using 5.34 in 5.31 gives

$$\frac{y^n}{1 - n} \frac{dv}{dx} + P(x)y = Q(x)y^n$$

Then multiply through by $\frac{(1 - n)}{y^n}$

$$\frac{dv}{dx} + (1 - n)P(x) \frac{y}{y^n} = \frac{(1 - n)Q(x)y^n}{y^n}$$

or

$$\frac{dv}{dx} + (1 - n)P(x)y^{1-n} = (1 - n)Q(x)$$

$$\frac{dv}{dx} + P_1(x)v = Q_1(x) \quad (5.35)$$

Where $P_1(x) = (1 - n)P(x)$ and $Q_1(x) = (1 - n)Q(x)$

□

Example 1

Determine the solution to

$$\frac{dy}{dx} + y = xy^3 \quad (5.36)$$

Solution:

This is a Bernoulli equation with $n = 3$ ¹⁶.

Let

$$v = y^{-2} \quad (5.37)$$

Then we differentiate

$$\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx} \quad (5.38)$$

¹⁴In 5.33, we are using implicit differentiation

¹⁵Laws of indices

¹⁶What is telling us is y^3

$$\begin{aligned}
-\frac{1}{2y^{-3}} \frac{dv}{dx} &= \frac{dy}{dx} \\
-\frac{y^3}{2} \frac{dv}{dx} &= \frac{dy}{dx}
\end{aligned} \tag{5.39}$$

Use 5.39 in 5.36 we have ,

$$-\frac{y^3}{2} \frac{dv}{dx} + y = xy^3$$

Multiplying through by $\frac{-2}{y^3}$

$$\begin{aligned}
\frac{dv}{dx} + -\frac{2}{y^3}y &= \frac{-2}{y^3}xy^3 \\
\frac{dv}{dx} - \frac{2}{y^2} &= -2x \\
\frac{dv}{dx} - 2(y^{-2}) &= -2x \\
\frac{dv}{dx} - 2v &= -2x
\end{aligned} \tag{5.40}$$

I.f

$$= e^{\int -2dx} = e^{-2x} \tag{5.41}$$

Multiplying 5.40 through by 5.41

$$\begin{aligned}
e^{-2x} \frac{dv}{dx} - 2ve^{-2x} &= -2xe^{-2x} \\
\frac{d}{dx} [ve^{-2x}] &= -2xe^{-2x}
\end{aligned}$$

Intergrate

$$ve^{-2x} = \int -2xe^{-2x} dx$$

Integration by parts

Let $u = -2x$, $\frac{du}{dx} = -2$

$$\frac{dv}{dx} = e^{-2x} \implies v = \frac{-1}{2}e^{-2x}$$

$$\begin{aligned}
ve^{-2x} &= -2x \cdot \frac{1}{-2}e^{-2x} - \int (-2) \left(-\frac{1}{2}e^{-2x} \right) dx \\
ve^{-2x} &= xe^{-2x} - \int e^{-2x} dx
\end{aligned}$$

$$ve^{-2x} = xe^{-2x} + \frac{1}{2}e^{-2x} + C$$

or

$$y^{-2}e^{-2x} = xe^{-2x} + \frac{1}{2}e^{-2x} + C$$

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CHAPTER 6

HIGHER ORDER EQUATIONS

Consider the differential equation of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-2} \frac{d^2 y}{dx^2} + a_{n-1} \frac{dy}{dx} + a_n y = F(x) \quad (6.1)$$

Where a_i 's are constants and $F(x)$ is a function of x , if $F(x) = 0$, we have

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-2} \frac{d^2 y}{dx^2} + a_{n-1} \frac{dy}{dx} + a_n y = 0 \quad (6.2)$$

Equation 6.2 is a homogeneous equation with constant coefficients.

Our task therefore is to Determine the function y satisfying equation 6.2. This function must be such that, all its derivatives are scalar multiples of each other.

The only function satisfying this condition is $y = e^{mx}$. Since

$$\frac{dy}{dx} = m e^{mx} \quad (6.3)$$

$$\frac{d^2 y}{dx^2} = m^2 e^{mx} \quad (6.4)$$

$$\vdots \quad (6.5)$$

$$\frac{d^n y}{dx^n} = m^n e^{mx} \quad (6.6)$$

Substituting equation 6.3 in equation 6.1 gives,

$$a_0 m^n e^{mx} + a_1 m^{n-1} e^{mx} + \dots + a_{n-2} m^2 e^{mx} + a_{n-1} m e^{mx} + a_n e^{mx} = 0$$

or

$$e^{mx} [a_0 m^n + a_1 m^{n-1} + \dots + a_{n-2} m^2 + a_{n-1} m + a_n] = 0$$

For $e^{mx} \neq 0$, we have

$$a_0 m^n + a_1 m^{n-1} + \dots + a_{n-2} m^2 + a_{n-1} m + a_n = 0 \quad (6.7)$$

Equation 6.7 is a polynomial of degree n in m . This equation is known as, The auxilliary/ the Characteristic equation of 6.2

The polynomial has n roots that are either

1. Real and distinct
2. Real and repeated
3. Complex conjugate roots

6.0.1 Real and Distinct Roots

If the roots of equation 6.7 are real and distinct, say

$$m = m_1, m_2, \dots, m_n$$

we have n fundamental/basic solutions

$$y_1 = e^{m_1 x}, y_2 = e^{m_2 x}, \dots, y_n = e^{m_n x}$$

and the general solution is

$$y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$$

Example

Determine the general solution to the equation

$$\frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 35y = 0$$

Solution

Auxilliary equation:

$$m^2 - 12m + 35 = 0$$

Which on solving

$$m_1 = 5, m_2 = 7$$

Fundamental Solution:

$$y_1 = e^{5x}, y_2 = e^{7x}$$

General Solution:

$$y = c_1 e^{5x} + c_2 e^{7x}$$

6.0.2 Real Repeated Roots

If the roots of the auxiliary equation are real and repeated i.e.

$$m_1 = m_2 = \dots = m_n = n$$

then, there is only one fundamental solution

$$y_i = e^{mx}, i = 1, 2, \dots, n$$

and the required general equation is

$$y = c_1 e^{mx} + c_2 x e^{mx} + c_3 x^2 e^{mx} + \dots + c_n x^{n-1} e^{mx}$$

Example

Determine the general solution to the equation

$$\frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 35y = 0$$

Solution

A|E:

$$m^2 - 12m + 35 = 0$$

Which on solving

$$m_1 = 5, m_2 = 7$$

Fundamental Solution:

$$y_1 = e^{5x}, y_2 = e^{7x}$$

General Solution:

$$y = c_1 e^{5x} + c_2 e^{7x}$$

Example

Determine the general equation solution to

$$\frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + y = 0$$

$$A : E ::> m^3 + 3m^2 + 3m + 1 = 0$$

$$f(m) = m^3 + 3m^2 + 3m + 1$$

$$f(1) = 1 + 3 + 3 + 1 = 8$$

$$f(-1) = -1 + 3 - 3 + 1 = 0$$

$\rightarrow m + 1$ is a factor

$$\begin{aligned}
 m^3 &= 3m_2^2m + 1 = (m+1)(m_2^2m_1) = 0 \\
 (m+1)(m+1)(m+1) &= 0 \\
 m &= -1(\text{thrice})
 \end{aligned}$$

Fundamental Solution :

$$y = e^{-x}$$

General Solution

$$y = c_1e^{-x} + c_2xe^{-x} + c_3x^2e^{-x}$$

6.0.3 Complex Conjugate Roots

Consider the 2^{nd} order differential equation given by

$$a_0 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_2y = 0$$

Whose Characteristic equation is

$$a_0m^2 + a_1m + a_2 = 0$$

Suppose the roots to this equation are

$$m = a \pm ib$$

That is $m_1 = a + ib, m_2 = a - ib$

The required general solution is

$$\begin{aligned}
 y &= c_1e^{(a-ib)x} + c_2e^{(a+ib)x} \\
 y &= c_1e^{ax} \cdot e^{-ibx} + c_2e^{ax} \cdot e^{ibx} \\
 y &= e^{ax} [c_1e^{-ibx} + c_2e^{ibx}]
 \end{aligned}$$

$$\text{But } \left. \begin{aligned} e^{-i\theta} &= \cos \theta - i \sin \theta \\ e^{i\theta} &= \cos \theta + i \sin \theta \end{aligned} \right\} \text{Euler's Formula.}$$

Hence we have

$$\begin{aligned}
 y &= e^{ax} [c_1(\cos bx - i \sin bx) + c_2(\cos bx + i \sin bx)] \\
 &= e^{ax} [(c_1 + c_2) \cos bx + i(c_2 - c_1) \sin bx] \\
 \text{Let } (c_1 + c_2) &= C_1 \\
 i(c_2 - c_1) &= C_2 \\
 \therefore y &= e^{ax} [C_1 \cos bx + C_2 \sin bx]
 \end{aligned}$$

, the required solution, a is the real part, b is the imaginary part.

Example

Determine the general solution to the equation

$$\frac{d^2y}{dx^2} + y \frac{dy}{dx} + 25y = 0$$

Solution

AE

$$\begin{aligned} m^2 + 6m + 25 &= 0 \\ m &= \frac{-6 \pm \sqrt{36 - 100}}{2} \\ &= \frac{-6 \pm \sqrt{-64}}{2} \\ &= \frac{-6 \pm 8i}{2} \\ m &= -3 \pm 4i \\ a &= -3, b = 4 \\ y &= e^{-3x} [\cos 4x + \sin 4x] \end{aligned}$$

Example

$$\frac{d^2y}{dx^2} + y = 0$$

Solution

A.E:

$$\begin{aligned} m^2 + 1 &= 0 \\ m^2 &= -1 \\ m &= \pm 1 \\ a &= 0, b = 1 \\ y &= e^0 [\cos x + \sin x] \\ &= e^0 [c_1 \cos x + c_2 \sin x] \\ y &= c_1 \cos x + c_2 \sin x \end{aligned}$$

Check:

$$\begin{aligned} y' &= -c_1 \sin x + c_2 \cos x \\ y'' &= -c_1 \cos x - c_2 \sin x \\ -c_1 \cos x - c_2 \sin x + c_1 \cos x + c_2 \sin x &= 0 \end{aligned}$$

6.1 Non-Homogeneous Equations

Consider the 2^{nd} order equation

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = F(x) \quad (6.8)$$

The equation has 2 solutions, namely

- i) The complementary function y_c
- ii) The particular integral y_p

6.1.1 The complementary Function y_c

This is the solution associated with the homogeneous part of 6.8. That is, the solution to

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (6.9)$$

6.1.2 The Particular Integral y_p

This is the solution associated with the non-homogeneous part, $F(x)$, and the required general equation is

$$y = y_c + y_p$$

CRAMER'S RULE

It is used to solve a system of linear equations.

$$\begin{aligned}x + y &= 40 \\2x + 4y &= 100 \\ \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 40 \\ 100 \end{pmatrix} \\ A = \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix} &\Rightarrow \begin{vmatrix} 1 & 1 \\ 2 & 4 \end{vmatrix} = 4 - 2 = 2 \\ A_x = \begin{pmatrix} 40 & 1 \\ 100 & 4 \end{pmatrix} &\Rightarrow \begin{vmatrix} 40 & 1 \\ 100 & 4 \end{vmatrix} = 160 - 100 = 60 \\ A_y = \begin{pmatrix} 1 & 40 \\ 2 & 100 \end{pmatrix} &\Rightarrow \begin{vmatrix} 1 & 40 \\ 2 & 100 \end{vmatrix} = 100 - 80 = 20 \\ x = \frac{|A_x|}{|A|} &= \frac{60}{2} = 30 \\ y = \frac{|A_y|}{|A|} &= \frac{20}{2} = 10\end{aligned}$$

7.1 The Method of Variation of Parameter

Consider the 2^{nd} order differential equation

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = F(x) \quad (7.1)$$

To determine the complementary function y_c , we solve the equation

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \quad (7.2)$$

Suppose, the solution to equation 7.2 is

$$y_c = c_1 y_1 + c_2 y_2 \quad (7.3)$$

By the method of variation of parameter, we take a particular integral

$$y + p = v_1 y_1 + v_2 y_2 \quad (7.4)$$

Where v_1 and v_2 are functions of x .

Differentiating equation 7.4 with respect to x gives

$$y'_p = v'_1 y_1 + v_1 y'_1 + v'_2 y_2 + v_2 y'_2$$

By Product rule

then, impose the condition

$$v'_1 y_1 + v'_2 y_2 = 0 \quad (7.5)$$

Such that

$$y'_p = v_1 y'_1 + v_2 y'_2 \quad (7.6)$$

Then differentiating equation 7.6 with respect to x , gives

$$y_p'' = v_1' y_1' + v_1 y_1'' + v_2' y_2' + v_2 y_2'' \quad (7.7)$$

Substituting 7.7, 7.6 and 7.4 in 7.1 gives

$$a_0 [v_1' y_1' + v_1 y_1'' + v_2' y_2' + v_2 y_2''] + a_1 [v_1 y_1' + v_2 y_2'] + a_2 [v_1 y_1 + v_2 y_2] = F(x)$$

or

$$v_1 [a_0 y_1'' + a_1 y_1' + a_2 y_1] + v_2 [a_0 y_2'' + a_1 y_2' + a_2 y_2] + a_0 [v_1' y_1' + v_2' y_2'] = F(x)$$

The 1st and 2nd terms vanish because y_1 and y_2 are solutions to equation 7.2.

Therefore, we have

$$v_1' y_1' + v_2' y_2' = \frac{F(x)}{a_0} \quad (7.8)$$

We then solve equation 7.5 and 7.8 using the Crammer's Rule.

$$\begin{aligned} v_1' y_1 + v_2' y_2 &= 0 \\ v_1' y_1' + v_2' y_2' &= \frac{F(x)}{a_0} \end{aligned}$$

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} v_1' \\ v_2' \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{F(x)}{a_0} \end{pmatrix}$$

$$v_1' = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{F(x)}{a_0} & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} \quad (7.9)$$

and

$$v_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & \frac{F(x)}{a_0} \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}} \quad (7.10)$$

Integrating equations 7.9 and 7.10, we obtain v_1 and v_2 respectively. Which we substitute to equation 7.4 to obtain the particular integral.

Example

Use the method of variation of parameters to determine the solution to

$$\frac{d^2 y}{dx^2} + y = \tan x \quad (7.11)$$

Solution

We first solve the equation

$$\frac{d^2 y}{dx^2} + y = 0 \quad (7.12)$$

Auxilliary equation :

$$\begin{aligned} m^2 + 1 &= 0 \\ m &= \pm i \end{aligned}$$

Therefore

$$y_c = c_1 \cos x + c_2 \sin x \quad (7.13)$$

By the method of variation of parameters, we take

$$y_p = v_1 \cos x + v_2 \sin x \quad (7.14)$$

Differentiating equation 7.14 w.r.t x gives

$$y'_p = v'_1 \cos x - v_1 \sin x + v'_2 \sin x + v_2 \cos x$$

Then impose the condition

$$v'_1 \cos x + v'_2 \sin x = 0 \quad (7.15)$$

Such that

$$y'_p = -v_1 \sin x + v_2 \cos x \quad (7.16)$$

Differentiating 7.16 with respect to x gives;

$$y''_p = -v'_1 \sin x - v_1 \cos x + v'_2 \cos x - v_2 \sin x \quad (7.17)$$

Substituting 7.17 and 7.14 in 7.11 gives

$$-v'_1 \sin x - v_1 \cos x + v'_2 \cos x - v_2 \sin x + v_1 \cos x + v_2 \sin x = \tan x$$

$$-v'_1 \sin x + v'_2 \cos x = \tan x \quad (7.18)$$

We now solve, equation 7.15 and 7.18 using Crammer's rule.

$$\begin{aligned}
-v'_1 \cos x + v'_2 \sin x &= 0 \\
-v'_1 \sin x + v'_2 \cos x &= \tan x \\
\begin{pmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{pmatrix} \begin{pmatrix} v'_1 \\ v'_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ \tan x \end{pmatrix} \\
v'_1 &= \frac{\begin{vmatrix} 0 & \sin x \\ \tan x & \cos x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} = \frac{-\sin x \tan x}{\cos^2 x + \sin^2 x} \\
&= -\sin x \tan x \\
&= -\sin x \frac{\sin x}{\cos x} \\
&= \frac{-\sin^2 x}{\cos x} \\
&= -\frac{1 - \cos^2 x}{\cos x} \\
&= \frac{-1}{\cos x} + \cos x \\
&= -\sec x + \cos x
\end{aligned}$$

From

$$\begin{aligned}
v'_1 &= -\sec x + \cos x \\
v_1 &= -\ln(\sec x + \tan x) + \sin x + c_3 \\
v'_2 &= \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & \tan x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} = \frac{\cos x \tan x}{\cos^2 x + \sin^2 x} \\
&= \frac{\cos x \tan x}{1} \\
&= \cos x \cdot \frac{\sin x}{\cos x} \\
&= \sin x
\end{aligned}$$

Therefore

$$\begin{aligned}
v'_2 &= \sin x \\
v_2 &= -\cos x + c_4 \text{ On integration}
\end{aligned}$$

Using v_1 and v_2 in equation 7.14 gives

$$y_p = [-\ln(\sec x + \tan x) + \sin x + c_3] \cos x + [-\cos x + c_4] \sin x$$

$$\begin{aligned}\therefore y_p &= -\cos x \ln(\sec x + \tan x) + \cos x \sin x + c_3 \cos x + -\cos x \sin x + c_4 \sin x \\ y_p &= -\cos x \ln(\sec x + \tan x) + c_3 \cos x + c_4 \sin x\end{aligned}$$

Required solution

$$y = y_c + y_p$$

$$\begin{aligned}y &= c_1 \cos x + c_2 \sin x - \cos x \ln(\sec x + \tan x) + c_3 \cos x + c_4 \sin x \\ y &= (c_1 + c_3) \cos x + (c_2 + c_4) \sin x - \cos x \ln(\sec x + \tan x) \\ y &= c_1 \cos x + c_2 \sin x - \cos x \ln(\sec x + \tan x) \\ C_1 &= c_1 + c_3 \\ C_2 &= c_2 + c_4\end{aligned}$$

Example

Use the method of variation of parameters

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = \frac{e^{-3x}}{x^3} \quad (7.19)$$

Solution: Solve

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0 \quad (7.20)$$

$$y_1 = c_1 e^{-3x} + c_2 x e^{-3x} \quad (7.21)$$

Auxilliary Equation:

$$\begin{aligned}m^2 + 6m + 9 &= 0 \\ (m + 3)(m + 3) &= 0 \\ (m + 3) \text{ twice} &= 0 \\ m &= -3 \\ y_c &= c_1 e^{-3x} + c_2 x e^{-3x}\end{aligned} \quad (7.22)$$

We take a solution of the form

$$y_p = v_1 e^{-3x} + v_2 x e^{-3x} \quad (7.23)$$

Differentiation equation 7.23 with respect to x gives

$$y'_p = v'_1 e^{-3x} - 3x v_1 e^{-3x} + v'_2 x e^{-3x} + v_2 e^{-3x} - 3v_2 x e^{-3x}$$

Impose

$$v_1' e^{-3x} + v_2' x e^{-3x} = 0$$

$$v_1' + x v_2' = 0 \quad (7.24)$$

$$y_p' = -3v_1 e^{-3x} + v_2 e^{-3x} - 3v_2 x e^{-3x} \quad (7.25)$$

$$y_p'' = -3v_1' e^{-3x} + 9v_1 e^{-3x} + v_2' e^{-3x} - 3v_2 e^{-3x} - 3v_2' x e^{-3x} - 3v_2 e^{-3x} + 9v_2 x e^{-3x} \quad (7.26)$$

Then use, 7.26, 7.25 and 7.23 in 7.19

$$\begin{aligned} & -3v_1' e^{-3x} + 9v_1 e^{-3x} + v_2' e^{-3x} - 3v_2 e^{-3x} - 3v_2' x e^{-3x} - 3v_2 e^{-3x} + 9v_2 x e^{-3x} \\ & + 6(-3v_1 e^{-3x} + v_2 e^{-3x} - 3v_2 x e^{-3x}) + 9(v_1 e^{-3x} + v_2 x e^{-3x}) = \frac{e^{-3x}}{x^3} \end{aligned}$$

$$-3v_1' e^{-3x} + v_2' e^{-3x} - 3v_2' x e^{-3x} = \frac{e^{-3x}}{x^3}$$

$$-3v_1' + v_2' - 3x v_2' = \frac{1}{x^3}$$

$$-3v_1' + v_2'(1 - 3x) = \frac{1}{x^3} \quad (7.27)$$

Use Crammer's rule to solve 7.24 and 7.27

$$\begin{aligned}
v_1' + xv_2' &= 0 \\
-3v_1' + (1-3x)v_2' &= \frac{1}{x^3} \\
\begin{pmatrix} 1 & x \\ -3 & 1-3x \end{pmatrix} \begin{pmatrix} v_1' \\ v_2' \end{pmatrix} &= \begin{pmatrix} 0 \\ \frac{1}{x^3} \end{pmatrix} \\
v_1' &= \frac{\begin{vmatrix} 0 & x \\ \frac{1}{x^3} & 1-3x \end{vmatrix}}{\begin{vmatrix} 1 & x \\ -3 & 1-3x \end{vmatrix}} = \frac{\frac{-1}{x^2}}{1-3x+3x} = -\frac{1}{x^2} \\
v_1 &= -\int \frac{1}{x^2} dx \\
&= -\int x^{-2} dx \\
&= \frac{1}{x} + c_3 \\
v_2' &= \frac{\begin{vmatrix} 1 & 0 \\ -3 & \frac{1}{x^3} \end{vmatrix}}{\begin{vmatrix} 1 & x \\ -3 & 1-3x \end{vmatrix}} = \frac{\frac{1}{x^3}}{1-3x+3x} = \frac{1}{x^3} \\
v_2 &= \int \frac{1}{x^3} dx \\
&= \int x^{-3} dx \\
&= -\frac{1}{2x^2} + c_4
\end{aligned}$$

7.2 The Method of Undetermined Coefficients

To determine the particular integral by the method of undetermined coefficient, the type of solution to be assumed depends of the function

$$F(x)$$

The tale below, gives common types of $F(x)$ and the type of solutions to be assumed.

$F(x)$	Solution to be Assumed
A	A , a constant
x	$Ax + B$, A and B are constants
x^2	$Ax^2 + Bx + C$, A, B, C are constants etc
e^{mx}	Ae^{mx}
$\cos ax$	$A \cos ax + B \sin ax$, A and B are constants
$\sin ax$	$A \cos ax + B \sin ax$, A and B are constants
$\cos ax \sin x$	$A \cos ax + B \sin ax$, A and B are constants

Example

Solve

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4e^{3x} \quad (7.28)$$

Solution:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$$

$$A : E | m^2 - 3m + 2 = 0$$

$$(m - 2)(m - 1) = 0$$

$$m_1 = 2$$

$$m_2 = 1$$

$$y_c = c_1 e^x + c_2 e^{2x}$$

$$F(x) = e^{3x}$$

Let

$$y_p = Ae^{3x}, \text{ where } A \text{ is a constant to be determined}$$

$$\left. \begin{array}{l} y_p' = 3Ae^{3x} \\ y_p'' = 9Ae^{3x} \end{array} \right\} \text{ use } \approx \text{ in 7.28}$$

$$9Ae^{3x} - 3(3Ae^{3x}) + 2Ae^{3x} = 4e^{3x}$$

$$2Ae^{3x} = 4e^{3x}$$

$$A = 2$$

$$y_p = 2e^{3x}$$

$$y = y_c + y_p$$

$$= c_1 e^x + c_2 e^{2x} + 2e^{3x}$$

Example

Determine the general solution to the equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 6e^{2x} \quad (7.29)$$

Solution:

A:E|

$$m^2 - 4y + 4 = 0$$

$$(m - 2)^2 = 0$$

$$m = 2(\text{twice})$$

$$y_c = c_1 e^{2x} + c_2 x e^{2x}$$

$$F(x) = e^{2x}$$

Since e^{2x} is in both y_c

$$e^{2x} \rightarrow x e^{2x} \rightarrow x^2 e^{2x}$$

$$y_p = A x^2 e^{2x} \quad (7.30)$$

$$y'_p = 2A x e^{2x} + 2A x^2 e^{2x} \quad (7.31)$$

$$y''_p = 2A e^{2x} + 8A x e^{2x} + 4A x^2 e^{2x} \quad (7.32)$$

Use 7.30, 7.31 and 7.32 in 7.29

$$2A e^{2x} + 8A x e^{2x} + 4A x^2 e^{2x} - 4A x e^{2x} - 8A x^2 e^{2x} + 4A x^2 e^{2x} = 6e^{2x}$$

$$2A e^{2x} = 6e^{2x}$$

$$A = 3$$

$$\therefore y_p = 3x^2 e^{2x}$$

$$Y = y_c + y_p$$

$$= c_1 e^{2x} + c_2 x e^{2x} + 3x^2 e^{2x}$$

Example

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 4x^2 \quad (7.33)$$

Solution First solve the complementary equation

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0 \quad (7.34)$$

$$m^2 - 3m + 2 = 0$$

$$(m - 1)(m - 2) = 0$$

$$m_1 = 1, m_2 = 2$$

$$y_c = c_1 e^x + c_2 e^{2x} \quad (7.35)$$

$F(x) = x^2$ Thus let

$$y_p = A x^2 + B x + c$$

Where A, B and C are constants to be determined.

$$\begin{aligned}y_p' &= 2Ax + B \\y_p'' &= 2A\end{aligned}$$

Substituting this in 7.33, gives

$$2A - 3(2Ax + B) + 2(Ax^2 + Bx + c) = 4x^2$$

$$2A - 6Ax - 3B + 2Ax^2 + 2Bx + 2C = 4x^2$$

$$x^0 : 2A - 3B + 2C = 0$$

$$x^1 : -6A + 2B = 0 \Rightarrow 2B = 6A, B = 3A$$

$$x^2 : 2A = 4 \Rightarrow A = 2$$

$$B = 2(2) = 4$$

$$2(2) - 3(4) + 2c = 0$$

$$4 - 12 + 2c = 0$$

$$2c = 8$$

$$c = 4$$

$$y_p = 2x^2 + 4x + 4$$

$$y = y_c + y_p$$

$$y = c_1 e^x + c_2 e^{2x} + 2x^2 + 4x + 4$$

Example

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 4y = 16x + 20e^x \quad (7.36)$$

Solution:

First solve

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 4y = 0 \quad (7.37)$$

$$m^2 + 5m + 4 = 0$$

$$(m + 1)(m + 4) = 0$$

$$m_1 = -1, m_2 = -4$$

$$y_c = c_1 e^{-x} + c_2 e^{-4x} \quad (7.38)$$

$$F(x) = (x, e^x)$$

$$\begin{aligned}
 y_p &= Ax + b + ce^x \\
 y'_p &= A + ce^x \\
 y''_p &= A + ce^x
 \end{aligned} \tag{7.39}$$

Substitue 7.39 in 7.36 to have

$$ce^x + 5(A + ce^x) + 4(Ax + B + ce^x) = 16x + 20e^x$$

$$ce^x + 5A + 5ce^x + 4Ax + 4B + 4ce^x = 16x + 20e^x$$

$$x^0 : 5A + 4B = 0$$

$$x^1 : 4A = 16, A = 4, B = -5$$

$$e^x : c + 5c + 4c = 20, c = 2$$

$$\therefore y_p = 4x - 5 + 2e^x$$

Hence

$$y = c_1e^{-x} + c_2e^{-4x} + 4x - 5 + 2e^x$$

$$y = y_c + y_p$$

CHAPTER 8

SYSTEMS OF EQUATIONS

Consider a system of linear equations in two dependent variables, x and y , and one independent variable t given as

$$a_0 \frac{dx}{dt} + a_1 \frac{dy}{dt} + a_2x + a_3y = f_1(t)$$

$$b_0 \frac{dx}{dt} + b_1 \frac{dy}{dt} + b_2x + b_3y = f_2(t)$$

Define the Operator

$$D = \frac{d}{dt}$$

Such that the system can be written as

$$a_0Dx + a_1Dy + a_2x + a_3y = f_1(t)$$

$$b_0Dx + b_1Dy + b_2x + b_3y = f_2(t)$$

$$(A_0D + a_2)x + (a_1D + a_3)y = f_1(t)$$

$$(b_0D + b_2)x + (b_1D + b_3)y = f_2(t)$$

We can now solve this system for x and y by elimination.

Example

Solve the following systems.

$$2 \frac{dx}{dt} - 2 \frac{dy}{dt} - 3x = t$$

$$2 \frac{dx}{dt} + 2 \frac{dy}{dt} + 3x + 8y = 2$$

Solution:

Letting $D = \frac{d}{dt}$, our system becomes

$$\begin{aligned} 2Dx - 2Dy - 3x &= t \\ 2Dx + 2Dy + 3x + 8y &= 2 \end{aligned}$$

or

$$\begin{aligned} (2D - 3)x - 2Dy &= t \\ (2D + 3)x + (2D + 8)y &= 2 \end{aligned}$$

To solve for x , we need to eliminate y by multiplying the first equation by $(2D + 8)$ and the second equation by $2D$ then add.

Thus we have

$$\begin{aligned} (2D + 8)(2D - 3)x - (2D + 8)(2Dy) &= (2D + 8)t \\ 2D(2D + 3)x + 2D(2D + 8)y &= 2D(2) \end{aligned}$$

or

$$\begin{aligned} [(2D + 8)(2D - 3) + 2D(2D + 3)]x &= 2 + 8t + 0 \\ [4D^2 - 6D + 16D - 24 + 4D^2 + 6D]x &= 2 + 8t \\ (8D^2 + 16D - 24)x &= 2 + 8t \end{aligned}$$

or

$$8\frac{d^2x}{dt^2} + 16\frac{dx}{dt} - 24x = 2 + 8t \quad (8.1)$$

First Solve

$$8m^2 + 16m - 24 = 0 \quad (8.2)$$

$$\begin{aligned} m^2 + 2m - 3 &= 0 \\ (m - 1)(m + 3) &= 0 \\ m = 1, m &= -3 \end{aligned}$$

$$x_c = c_1e^{-3t} + c_2e^t \quad (8.3)$$

$$\begin{aligned} F(t) &= 2 + 8t \\ x_p &= At + B \end{aligned}$$

$$\begin{aligned} x'_p &= A \\ x''_p &= 0 \end{aligned} \quad (8.4)$$

Substituting back 8.4 in 8.1

$$8(0) + 16A - 24(At + B) = 2 + 8t$$

$$16A - 24(At + b) = 2 + 8t$$

$$16A - 24At - 24B = 2 + 8t$$

$$t^0 : 16A - 24B = 2$$

$$t^1 : -24A = 8$$

$$A = -\frac{1}{3}$$

$$B = -\frac{11}{36}$$

$$x_p = -\frac{1}{3}t - \frac{11}{36}$$

$$x = c_1e^{-3t} + c_2e^t - \frac{1}{3}t - \frac{11}{36}$$

$$x = x_c + x_p$$

$$= c_1e^{-3t} + c_2e^t - \frac{1}{3}t - \frac{11}{36}$$

$$2\frac{dx}{dt} - 2\frac{dy}{dt} - 3x = t$$

$$2\frac{dx}{dt} + 2\frac{dy}{dt} + 3x + 8y = 2$$

To determine the remaining variable (y), we eliminate the derivatives of y from the given system.

$\therefore$ adding the two equations gives,

$$4\frac{dx}{dt} + 8y = t + 2$$

$$8y = t + 2 - 4\frac{dx}{dt}$$

$$y = \frac{1}{8} \left[t + 2 - 4\frac{d}{dt} \left(c_1e^{-3t} + c_2e^t - \frac{1}{3}t - \frac{11}{36} \right) \right]$$

$$= \frac{1}{8} \left[t + 2 - 4 \left(-3c_1e^{-3t} + c_2e^t - \frac{1}{3} \right) \right]$$

$$= \frac{1}{8} \left[t + 2 - (-12c_1e^{-3t} + 4c_2e^t + \frac{4}{3}) \right]$$

$$= \frac{1}{8} \left[t + 12c_1e^{-3t} - 4c_2e^t + \frac{10}{3} \right]$$

$$= \frac{1}{8}t + \frac{3}{2}c_1e^{-3t} - \frac{1}{2}c_2e^t + \frac{5}{12}$$

CHAPTER 9

EQUATIONS WITH VARIABLE COEFFICIENTS

Consider an equation of the form

$$a_0(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_2(x)y = F(x) \quad (9.1)$$

There exists no exact method of determining the solution to 9.1 but the solution can be approximated by power series, unless the equation is of a special kind e.g., the Cauchy-Euler which is the form

$$a_0x^2\frac{d^2y}{dx^2} + a_1x\frac{dy}{dx} + a_2y = F(x) \quad (9.2)$$

Note:

$x^n \frac{d^n y}{dx^n}$ is evident in each term.

Theorem 3: T

The substitution $x = e^t$ reduces the Cauchy-Euler 9.2 into a linear equation with constant coefficients.

Proof. Let

$$x = e^t \quad (9.3)$$

$$t = \ln(x)$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt} \quad (9.4)$$

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{d}{dx} \left[\frac{dy}{dx} \right] \\
&= \frac{d}{dx} \left[\frac{1}{x} \cdot \frac{dy}{dt} \right] \\
&= \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dx} \left[\frac{dy}{dt} \right] \\
&= \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dt} \left[\frac{dy}{dt} \frac{dt}{dx} \right] \\
&= \frac{-1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d^2y}{dt^2} \cdot \frac{1}{x} \\
&= \frac{1}{x^2} \left[\frac{d^2y}{dt^2} - \frac{dy}{dt} \right]
\end{aligned} \tag{9.5}$$

Substituting 9.3, 9.4 and 9.5 in 9.2 gives

$$\begin{aligned}
a_0 x^2 \left[\frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \right] + a_1 x \left[\frac{1}{x} \frac{dy}{dt} \right] + a_2 y &= F(e^t) \\
a_0 \frac{d^2y}{dt^2} + (a_1 - a_0) \frac{dy}{dt} + a_2 y &= F(e^t)
\end{aligned}$$

Thus if

$$a_0 = b_0, a_1 = b_1, a_2 = b_2$$

we have

$$b_0 \frac{d^2y}{dt^2} + b_1 \frac{dy}{dt} + b_2 y = F(e^t) \tag{9.6}$$

Equation 9.6 is a linear equation in y and t with constant coefficients. $\square$

Example

Determine the general solution to the equation

$$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3 \tag{9.7}$$

Solution:

This is a Cauchy Euler equation of order 2. Let

$$x = e^t \rightarrow t = \ln x \tag{9.8}$$

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt} \tag{9.9}$$

and

$$\frac{d^2y}{dx^2} = \frac{1}{x^2} \left[\frac{d^2y}{dt^2} - \frac{dy}{dt} \right] \tag{9.10}$$

Substituting 9.8, 9.9 and 9.10 in 9.7 gives.

$$\begin{aligned}
 x^2 \left[\frac{1}{x^2} \left(\frac{d^2 y}{dx^2} - \frac{dy}{dt} \right) \right] - 2x \left(\frac{1}{x} \cdot \frac{dy}{dt} \right) + 2y &= e^{3t} \\
 \frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2 \frac{dy}{dt} + 2y &= e^{3t} \\
 \frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + 2y &= e^{3t}
 \end{aligned} \tag{9.11}$$

Auxilliary Equation:

$$\begin{aligned}
 m^2 - 3m + 2 &= 0 \\
 (m - 1)(m - 2) &= 0 \\
 m = 1, m = 2 \\
 y_c &= c_1 e^t + c_2 e^{2t} \\
 F(t) &= e^{3t} \\
 y_p &= A e^{3t} \\
 y'_p &= 3A e^{3t} \\
 y''_p &= 9A e^{3t}
 \end{aligned} \tag{9.12}$$

Use 9.12 in 9.11

$$\begin{aligned}
 9A e^{3t} - 9A e^{3t} + 2A e^{3t} &= e^{3t} \\
 e^{3t} : 2A = 1, A &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 y_p &= \frac{1}{2} e^{3t} \\
 \therefore y &= y_c + y_p \\
 &= c_1 - 1e^t + c_2 e^{2t} + \frac{1}{2} e^{3t} \\
 y &= c_1 x + c_2 x^2 + \frac{1}{2} x^3
 \end{aligned}$$

Example

Solve the equation

$$x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 2y = 4 \sin(\ln x) \tag{9.13}$$

Solution:

This is a Cauchy Euler equation of order 2

Let

$$x = e^t \rightarrow t = \ln x \tag{9.14}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \quad (9.15)$$

$$\frac{d^2y}{dx^2} = \frac{1}{x^2} \left[\frac{d^2y}{dt^2} - \frac{dy}{dt} \right] \quad (9.16)$$

$$x^2 \left[\frac{1}{x^2} \left(\frac{d^2y}{dx^2} - \frac{dy}{dt} \right) \right] - 5x \left[\frac{1}{x} \frac{dy}{dt} \right] + 2y = 4 \sin t$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dt} - 5 \frac{dy}{dt} + 2y = 4 \sin t$$

$$\frac{d^2y}{dx^2} - 6 \frac{dy}{dt} + 2y = 4 \sin t \quad (9.17)$$

$$m^2 - 6m + 2 = 0$$

$$m_1 = 3 + \sqrt{7}, m_2 = 3 - \sqrt{7}$$

$$y_c = c_1 e^{(3+\sqrt{7})t} + c_2 e^{(3-\sqrt{7})t}$$

$$F(t) = \sin t$$

$$y_p = A \cos t + B \sin t$$

$$y_p' = -A \sin t + B \cos t \quad (9.18)$$

$$y_p'' = -A \cos t - B \sin t$$

From 9.18

$$-A \cos t - B \sin t - 6(-A \sin t + B \cos t) + 2A \cos t + 2B \sin t$$

$$-A \cos t - B \sin t + 6A \sin t - 6B \cos t + 2A \cos t + 2B \sin t$$

$$\cos t : -A - 6B + 2A = 0$$

$$A - 6B = 0$$

$$\sin t : -B + 6A + 2B$$

$$6A + B = 4$$

$$A = \frac{24}{37}$$

$$B = \frac{4}{37}$$

CHAPTER 10

SERIES SOLUTIONS TO ODE'S

Consider the 2^{nd} order differential equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x)y = 0 \quad (10.1)$$

Dividing through by $a_0(x)$ gives

$$\frac{d^2 y}{dx^2} + \frac{a_1(x)}{a_0(x)} \frac{dy}{dx} + \frac{a_2(x)}{a_0(x)} y = 0$$

Setting

$$P(x) = \frac{a_1(x)}{a_0(x)}$$

and

$$Q(x) = \frac{a_2(x)}{a_0(x)}$$

we have

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0 \quad (10.2)$$

Definition 8. A point x_0 is said to be an ordinary point of equation 10.1 if both $P(x)$ and $Q(x)$ are defined at x_0

Example

Given

$$(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0$$

we divide through by $(x^2 - 1)$ to have

$$\frac{d^2 y}{dx^2} + \frac{3x}{x^2 - 1} \frac{dy}{dx} + \frac{x}{x^2 - 1} y = 0$$

$$\Rightarrow P(x) = \frac{3x}{x^2 - 1}, Q(x) = \frac{x}{x^2 - 1}$$

We note that both $P(x)$ and $Q(x)$ are defined at every other point except $x = -1$ and $x = 1$

Therefore, all points except -1 and 1 are ordinary points. -1 and 1 are called **singular points**. (Singular point is a point where both $P(x)$ and $Q(x)$ are not defined.)

If x_0 is a singular point but, $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are both defined at x_0 , then x_0 is called a regular singular point of equation 10.1.

Example

Given the equation

$$2x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0$$

$$P(x) = \frac{x}{2x^2} = \frac{1}{2x}$$

$$Q(x) = \frac{x^2 - 1}{2x^2}$$

Note 0 is a singular point of 10.1. Further $xP(x) = x \cdot \frac{1}{2x} = \frac{1}{2}$, $x^2Q(x) = x^2 \cdot \frac{(x^2-1)}{2x^2} = \frac{x^2-1}{2}$

Both defined at 0, hence 0 is a regular singular point of 1

10.1 CASE 1: Solution about an Ordinary Point

10.1.1 Taylor Series Solution

Recall: The Taylor series expansion of a function $y(x)$ about point x_0 (ordinary point) is given as

$$y(x) = y(x_0) + (x - x_0)y'(x_0) + (x - x_0)^2 \frac{y''(x_0)}{2!} + \dots \quad (10.3)$$

If $x_0 = 0$, we have

$$y(x) = y(0) + xy'(0) + \frac{x^2 y''(0)}{2!} + \frac{x^3 y'''(0)}{3!} + \dots \quad (10.4)$$

10.3 was referred as Taylor series expansion, 10.4 was referred as Macluris series expansion.

Example

Determine the Taylor's series solution of

$$(x^2 - 1) \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + xy = 0 \quad (A)$$

Given $y(0) = 4, y'(0) = 6$

$$x_0 = 0$$

From equation 10.4 we have

$$y(x) = y(0) + xy'(0) + x^2 \frac{y''(0)}{2!} + x^3 \frac{y'''(0)}{3!} + \dots$$

The given equation can be given as

$$\begin{aligned} \frac{d^2y}{dx^2} + \frac{3x}{x^2-1} \frac{dy}{dx} + \frac{x}{x^2-1} y &= 0 \\ \frac{d^2y}{dx^2} &= \frac{-3x}{x^2-1} \frac{dy}{dx} - \frac{x}{x^2-1} y = 0 \\ \left. \frac{d^2y}{dx^2} \right|_{x=0} &= \left. \frac{-3(0)}{0-1} \frac{dy}{dx} \right|_{x=0} - \frac{0}{0-1} y(0) \\ &= 0(6) - 0(4) \\ &= 0 \\ \Rightarrow y''(0) &= 0 \end{aligned}$$

Differentiation the given equation gives, A.

$$2x \frac{d^2y}{dx^2} + (x^2-1) \frac{d^3y}{dx^3} + 3 \frac{dy}{dx} + 3x \frac{d^2y}{dx^2} + y + x \frac{dy}{dx} = 0$$

By product rule

$$\begin{aligned} (x^2-1) \frac{d^3y}{dx^3} &= -2x \frac{d^2y}{dx^2} - 3 \frac{dy}{dx} - 3x \frac{d^2y}{dx^2} - y - x \frac{dy}{dx} \\ \frac{d^3y}{dx^3} &= \frac{-2x}{x^2-1} \frac{d^2y}{dx^2} - \frac{3}{x^2-1} \frac{dy}{dx} - \frac{3x}{x^2-1} \frac{d^2y}{dx^2} - \frac{y}{x^2-1} - \frac{x}{x^2-1} \frac{dy}{dx} \\ \left. \frac{d^3y}{dx^3} \right|_{x=0} &= \frac{-2(0)}{0-1} (0) - \frac{3}{0-1} (6) - \frac{3(0)}{0-1} (0) - \frac{4}{0-1} - \frac{0}{0-1} \\ &= +18 + 4 \\ &= 22 \end{aligned}$$

$$y(x) = y(0) + xy'(0) + x^2 \frac{y''(0)}{2!} + x^3 \frac{y'''(0)}{3!} + \dots$$

$$y(x) = 4 + 6(x) + x(0) + x^3 \cdot \frac{22}{6} + \dots$$

$$y(x) = 4 + 6x + \frac{11}{3}x^3$$

10.1.2 General Power Series

If x_0 is an ordinary point, we take a solution of the form

$$y = \sum_{n=0}^{\infty} C_n (x - x_0)^n$$

Differentiate twice, then substitute to the given equation.

Example

Determine the power series solution to the equation

$$\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 + 2)y = 0 \quad (10.5)$$

About the ordinary point $x_0 = 0$

Since 0 is an ordinary point, we take a solution

$$y = \sum_{n=0}^{\infty} C_n x^n \quad (10.6)$$

Differentiating equation 10.6 twice with respect to x gives

$$\frac{dy}{dx} = \sum_{n=1}^{\infty} n C_n x^{n-1} \quad (10.7)$$

and

$$\frac{d^2y}{dx^2} = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} \quad (10.8)$$

Substituting 10.6, 10.7 and 10.8 in 10.5 gives

$$\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + x \sum_{n=1}^{\infty} n C_n x^{n-1} + (x^2 + 2) \sum_{n=0}^{\infty} C_n x^n = 0$$

Or

$$\sum_{n=2}^{\infty} \underbrace{n(n-1) C_n x^{n-2}}_I + \sum_{n=1}^{\infty} n C_n x^n + \sum_{n=0}^{\infty} \underbrace{C_n x^{n+2}}_{II} + \sum_{n=0}^{\infty} 2 C_n x^n = 0 \quad (10.9)$$

To make the powers of x equal in equation 10.9. In

I) Let

$$\begin{aligned} S = 2 - 2, \Rightarrow n = 2 + s \therefore \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} \\ = \sum_{s=0}^{\infty} (s+2)(s+1) C_{s+2} x^s \\ = \sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} C^n \end{aligned} \quad (10.10)$$

Since s is a dummy variable.

II) Let

$$\begin{aligned}
 n+2=p, n=p-2 \therefore \sum_{n=0}^{\infty} C_n x^{n+2} \\
 = \sum_{p=2}^{\infty} C_{p-2} x^p \\
 = \sum_{n=2}^{\infty} C_{n-2} x^n
 \end{aligned} \tag{10.11}$$

Substituting 10.10, 10.11 in 10.9 gives

$$\sum_{n=0}^{\infty} (n+1)(n+1)C_{n+2}x^n + \sum_{n=1}^{\infty} nC_n x^n + \sum_{n=2}^{\infty} C_{n-2}x^n + \sum_{n=0}^{\infty} 2C_n x^n = 0 \tag{10.12}$$

We achieve, all powers are equal.

To make the range of $\sum$ equal in 10.12 we sum out for 0 and 1 in the first, second and forth terms. To have

$$\begin{aligned}
 2(1)C_2x^0 + (3)(2)C_3x^1 \\
 + \sum_{n=2}^{\infty} nC_n x^n + \sum_{n=2}^{\infty} C_{n-2}x^n \\
 + 2C_0x^0 + 2C_1 + \sum_{n=2}^{\infty} 2C_n x^n = 0
 \end{aligned}$$

or

$$2C_0 + 2C_2 + 3C_1x + 6C_3x + \sum_{n=2}^{\infty} [(n+2)C_{n+2} + (n+2)C_n + C_{n-2}] x^n = 0 \tag{10.13}$$

Equation coefficients, we have

$$\begin{aligned}
 x^0 : 2C_0 + 2C_2 = 0, C_2 = -C_0 \\
 x^1 : 3C_1 + 6C_3 = 0, C_1 + 2C_3 = 0, C_3 = -\frac{1}{2}C_1 \\
 x^n : (n+1)(n+1)C_{n+2} + (n+2)C_n + C_{n-2} = 0 \\
 (n+2)(n+1)C_{n+2} = -(n+2)C_n - C_{n-2} \\
 C_{n+2} = \frac{-(n+2)C_n - C_{n-2}}{(n+2)(n+1)}
 \end{aligned}$$

When

$$\begin{aligned}
 n = 2 : C_4 &= \frac{-(4)C_2 - C_0}{(4)(3)} = \frac{-4(-C_0) - C_0}{12} = \frac{4C_0 - C_0}{12} \\
 &= \frac{3C_0}{12} \\
 &= \frac{C_0}{4} \\
 n = 3 : C_5 &= \frac{-(5)C_3 - C_1}{(5)(4)} = \frac{-5\left(\frac{-C_1}{2}\right) - C_1}{20} \\
 &= \frac{\frac{5C_1}{2} - C_1}{20} \\
 &= \frac{3}{40}C_1
 \end{aligned}$$

Substituting these constants in the assumed solution gives

$$\begin{aligned}
 y &= \sum_{n=0}^{\infty} C_n x^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + \cdots \\
 &= C_0 + C_1 x - C_0 x^2 + -\frac{C_1}{2} x^3 + \frac{C_0}{4} x^4 + \frac{3}{40} C_1 x^5 + \cdots \\
 &= C_0 \left[1 - x^2 + \frac{1}{4} x^4 + \cdots \right] + C_1 \left[x - \frac{x^3}{2} + \frac{3}{40} x^5 + \cdots \right]
 \end{aligned}$$

Procedure

1. Identify the nature of point (ordinary) and assume the required solution, y
2. Compute the required derivatives and substitute (note the change of range)
3. Make the powers equal , (x^n)
4. Make the range of summation equal. (guided by the highest)
5. Equate coefficients to determine the constants.

10.2 CASE 2: Solution about a regular singular point

If x_0 is a regular singular point to the given differential equation, we take a solution of the form

$$y = \sum_{n=0}^{\infty} C_n x^{n+r}$$

Example

Determine the series solution to the differential equation

$$2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (x - 5)y = 0 \quad (10.14)$$

About the point $x_0 = 0$

Solution

Here

$$P(x) = \frac{-x}{2x^2} = \frac{1}{2x}$$

$$Q(x) = \frac{x-5}{2x^2}$$

Both undefined at $x = 0$, hence 0 is a singular point of 10.14.

Further ,

$$xP(x) = x \left(-\frac{1}{2x} \right) = -\frac{1}{2}$$

$$x^2Q(x) = x^2 \left(\frac{x-5}{2x^2} \right) = \frac{x-5}{2}$$

Both defined at $x = 0$, hence 0 is a regular singular point of 10.14

We therefore take a solution of

$$y = \sum_{n=0}^{\infty} C_n x^{n+r} \quad (10.15)$$

Differentiate twice with respect to x to have

$$\frac{dy}{dx} = \sum_{n=0}^{\infty} (n+r) C_n x^{n+r-1} \quad (10.16)$$

$$\frac{d^2y}{dx^2} = \sum_{n=0}^{\infty} (n+r)(n+r-1) C_n x^{n+r-2} \quad (10.17)$$

Note, in 10.16, 10.17, there is no change in the range.

Substituting 10.15, 10.16 and 10.17 in ?? gives

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1) C_n x^{n+r-2} - x \sum_{n=0}^{\infty} (n+r) C_n x^{n+r-1} + (x-5) \sum_{n=0}^{\infty} C_n x^{n+r} = 0 \quad (10.18)$$

or

$$\sum_{n=0}^{\infty} \underbrace{2(n+r)(n+r-1) C_n x^{n+r}}_I + \sum_{n=0}^{\infty} \underbrace{-(n+r) C_n x^{n+r}}_{II} + \sum_{n=0}^{\infty} \underbrace{C_n x^{n+r+1}}_{III} + \sum_{n=0}^{\infty} \underbrace{-5 C_n x^{n+r}}_{IV} = 0$$

In *III*, set

$$\begin{aligned}
 n+1 = s &\Rightarrow n = s-1 \Rightarrow \sum_{n=0}^{\infty} C_n x^{n+r+1} \\
 &= \sum_{s=1}^{\infty} C_{s-1} x^{3+r} \\
 &= \sum_{n=1}^{\infty} C_{n-1} x^{n+r}
 \end{aligned} \tag{10.19}$$

Substituting 10.19 in 10.18 gives

$$\sum_{n=0}^{\infty} 2(n+r)(n+r-1)C_n x^{n+r} + \sum_{n=0}^{\infty} -(n+r)C_n x^{n+r} + \sum_{n=1}^{\infty} C_{n-1} x^{n+r} + \sum_{n=0}^{\infty} -5C_n x^{n+r} = 0 \tag{10.20}$$

We now sum out 0 in the 1st, 2nd and 4th terms

$$2r(r-1)C_0 x^r + \sum_{n=1}^{\infty} 2(n+r)(n+r-1)C_n x^{n+r} - rC_0 x^r + \sum_{n=1}^{\infty} -(n+r)C_n x^{n+r} + \sum_{n=1}^{\infty} C_{n-1} x^{n+r} - 5C_0 x^r + \sum_{n=1}^{\infty} -5C_n x^{n+r} = 0$$

or

$$(2r^2 - 3r - 5)C_0 x^r + \sum_{n=1}^{\infty} \{[2(n+r)(n+r-1) - (n+r) - 5]C_n + C_{n-1}\} x^{n+r} = 0 \tag{10.21}$$

Equating coefficients to 0, we have

$$x^r : 2r^2 - 3r - 5 = 0 \tag{10.22}$$

Equation 10.22 is known as the indicial equation which on solving gives,

$$r_1 = -1, r_2 = \frac{5}{2}$$

$$x^{n+r} : [2(n+r)(n+r-1) - (n+r) - 5]C_n + C_{n-1} = 0 \tag{10.23}$$

When $r = \frac{5}{2}$

$$\left[2 \left(n + \frac{5}{2} \right) \left(n + \frac{5}{2} - 1 \right) - \left(n + \frac{5}{2} \right) - 5 \right] C_n + C_{n-1} = 0$$

$$\left[2n^2 + 8n + \frac{15}{2} - n - \frac{5}{2} - \frac{15}{2} \right] C_n + C_{n-1} = 0$$

$$(2n^2 + 7n)C_n + C_{n-1} = 0$$

$$C_n = \frac{-C_{n-1}}{n(2n+7)}, n = 1, 2, 3, \dots$$

When

$$\begin{aligned}
 n = 1, C_1 &= \frac{-C_0}{(1)(2+7)} = -\frac{1}{9}C_0 \\
 n = 2, C_2 &= \frac{-C_1}{2(4+7)} = \frac{-C_1}{22} = -\frac{1}{22} \left(\frac{-1}{9}C_0 \right) = \frac{1}{198}C_0 \\
 y &= \sum_{n=0}^{\infty} C_n x^{n+r} \\
 y &= \sum_{n=0}^{\infty} C_n x^{n+\frac{5}{2}} = C_0 x^{\frac{5}{2}} + C_1 x^{1+\frac{5}{2}} + C_2 x^{2+\frac{5}{2}} + \dots \\
 &= C_0 x^{\frac{5}{2}} - \frac{1}{9} C_0 x^{1+\frac{5}{2}} + \frac{1}{198} C_0 x^{2+\frac{5}{2}} + \dots \\
 &= C_0 x^{\frac{5}{2}} \left[1 - \frac{x}{9} + \frac{x^2}{198} + \dots \right]
 \end{aligned}$$

When

$$\begin{aligned}
 r &= -1 \\
 [2(n-1)(n-2) - (n-1) - 5] C_n + C_{n-1} &= 0 \\
 [2(n^2 - 3n + 2) - n + 1 - 5] C_n + C_{n-1} &= 0 \\
 [2n^2 - 6n + 4 - n + 1 - 5] C_n + C_{n-1} &= 0 \\
 (2n^2 - 7n) C_n + C_{n-1} &= 0 \\
 C_n &= \frac{-C_{n-1}}{(2n^2 - 7n)}, n = 1, 2, 3, \dots \\
 n = 1 : C_1 &= \frac{-C_0}{2-7} = \frac{-C_0}{-5} \\
 n = 2 : C_2 &= \frac{-C_1}{8-14} = \frac{-C_1}{-6} \\
 &= \frac{1}{6} C_1 \\
 &= \frac{1}{6} \left(\frac{-C_0}{-5} \right) \\
 &= \frac{1}{30} C_0 \\
 y &= \sum_{n=0}^{\infty} C_n x^{n-1} = C_0 x^{-1} + C_1 x^{1-1} + C_2 x^{2-1} + \dots \\
 &= C_0 x^{-1} + \frac{C_0}{5} + \frac{C_0}{30} x^1
 \end{aligned}$$

The Wronskian Method

When y_1 and y_2 are the two fundamental solutions of the homogeneous equation

$$y'' + Py' + Qy = 0$$

then the Wronskian $W(y_1, y_2)$ is the determinant of the matrix

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}$$

So

$$W(y_1, y_2) = y_1 y_2' - y_2 y_1'$$

Since y_1 and y_2 are linearly independent, the value of the Wronskian cannot equal zero.

The Particular Solution

Using the Wronskian we can now find the particular solution of the differential equation

$$y'' - Py' + Qy = F(x)$$

using the formula

$$y_p(x) = -y_1(x) \int \frac{y_2(x)F(x)}{W(y_1, y_2)} + y_2(x) \int \frac{y_1(x)F(x)}{W(y_1, y_2)}$$